

Pharmacotherapy 2: Ultimate Question Bank

Batch 4: Lecture 4 (Neuroleptics / Antipsychotics)

50 High-Yield Exam Questions with Clinical Explanations

101. Positive symptoms of Schizophrenia include:

A) Blunted affect and apathy

B) Hallucinations and Delusions

C) Social withdrawal

D) Cognitive impairment



Hint: Positive symptoms refer to added abnormal behaviors like hearing voices (auditory hallucinations) or having fixed false beliefs (delusions), such as believing one is controlled by an outside force.

102. Negative symptoms of Schizophrenia (like apathy and blunted affect) respond better to:

A) Typical antipsychotics (1st Gen)

B) Atypical antipsychotics (2nd Gen)

C) Benzodiazepines

D) Mood stabilizers



Hint: While Typical agents only treat positive symptoms effectively, Atypical agents (like Clozapine and Risperidone) ameliorate both positive and negative symptoms due to their dual mechanism.

103. Typical (First-Generation) antipsychotics primarily exert their main therapeutic effect by blocking which receptor?

A) Serotonin 5-HT_{2A}

B) Dopamine D₂

C) Histamine H₁

D) Muscarinic M₁



Hint: Typical neuroleptics are competitive inhibitors at the D₂ receptors in the brain, effectively reducing the excess dopamine activity that causes psychosis.

104. Psychosis is hypothesized to result from excessive Dopamine activity in which specific brain pathway?

- A) Nigrostriatal pathway
- B) Mesolimbic system**
- C) Tuberoinfundibular pathway
- D) Mesocortical pathway



Hint: The clinical efficacy of typical neuroleptics correlates closely with their ability to block D2 receptors specifically in the mesolimbic system.

105. Blocking Dopamine receptors in the Nigrostriatal pathway causes which unwanted effect?

- A) Therapeutic antipsychotic effect
- B) Extrapyramidal Symptoms (EPS) and Parkinson-like symptoms**
- C) Hyperprolactinemia
- D) Sedation



Hint: The doctor explained that the Nigrostriatal pathway controls voluntary movement. Blocking dopamine here mimics Parkinson's disease, causing tremors and rigidity (EPS).

106. Blocking Dopamine receptors in the Tuberoinfundibular pathway leads to:

- A) Extrapyramidal Symptoms (EPS)
- B) Hyperprolactinemia (elevated prolactin)**
- C) Worsening of negative symptoms
- D) Powerful anti-emetic effect



Hint: Dopamine normally inhibits prolactin release in the pituitary. Blocking D2 receptors removes this inhibition, causing prolactin levels to spike (leading to galactorrhea and gynecomastia).

107. Why do Atypical antipsychotics have a significantly lower incidence of EPS compared to Typicals?

A) They do not block Dopamine receptors at all

B) Their strong blockade of Serotonin (5-HT_{2A}) receptors modulates and reduces the risk of EPS

C) They block Muscarinic receptors exclusively

D) They are metabolized much faster



Hint: Atypicals are unique because they block 5-HT_{2A} receptors alongside D₂ receptors (or have higher affinity for D₄). This unique profile protects against severe motor side effects.

108. High potency typical antipsychotics, such as Haloperidol, are associated with a higher risk of:

A) Severe Sedation

B) Extrapyramidal Symptoms (EPS)

C) Orthostatic Hypotension

D) Strong Anticholinergic effects



Hint: Haloperidol has very low anticholinergic activity. Because it blocks dopamine without blocking acetylcholine, the resulting imbalance severely triggers EPS.

109. Why does Thioridazine (a typical antipsychotic) cause FEWER extrapyramidal disturbances compared to Haloperidol?

A) It is an atypical antipsychotic

B) It strongly dampens cholinergic activity (strong antimuscarinic effect), restoring the balance in the striatum

C) It does not cross the blood-brain barrier

D) It only blocks serotonin receptors



Hint: EPS is caused by a relative excess of acetylcholine when dopamine is blocked. Thioridazine also blocks acetylcholine (anticholinergic), restoring the 'seesaw' balance and minimizing EPS.

110. Antipsychotics are effective anti-emetics because they block D2-dopaminergic receptors in the:

- A) Vestibular apparatus
- B) Chemoreceptor Trigger Zone (CTZ) of the medulla**
- C) Vagus nerve terminals
- D) Cortex

 **Hint:** Most older neuroleptics (like prochlorperazine) are great for drug-induced nausea because they block the D2 receptors in the CTZ. (Note: Atypicals are NOT used as anti-emetics).

111. Acute Dystonia and other Extrapyrimal Symptoms (EPS) are effectively treated by administering:

- A) Dopamine agonists
- B) Anticholinergic drugs (e.g., Benztropine)**
- C) Beta-blockers
- D) Benzodiazepines

 **Hint:** To fix the drug-induced Parkinsonism (EPS), we must block the relatively hyperactive cholinergic system using an anticholinergic like Benztropine.

112. Tardive Dyskinesia, a late-onset motor disorder seen with chronic neuroleptic use, is believed to be caused by:

- A) Complete Dopamine depletion
- B) Compensatory up-regulation and supersensitivity of Dopamine receptors**
- C) Serotonin excess
- D) Acetylcholine deficiency

 **Hint:** Long-term D2 blockade makes the brain synthesize more receptors to compensate. The neurons become 'supersensitive' to dopamine, causing involuntary movements of the lips and tongue.

113. Which atypical antipsychotic has the lowest risk of causing Tardive Dyskinesia but carries a fatal risk of Agranulocytosis?

- A) Haloperidol
- B) Fluphenazine
- C) Clozapine**
- D) Chlorpromazine



Hint: Clozapine is superior for refractory schizophrenia and has very low EPS/TD risk, but it can wipe out white blood cells (Agranulocytosis in 1-2%), requiring strict blood monitoring.

114. Neuroleptic Malignant Syndrome (NMS) is a potentially fatal reaction that presents with severe muscle rigidity and:

- A) Severe Hypothermia
- B) High Fever (Hyperthermia)**
- C) Bradycardia
- D) Hypertension only



Hint: NMS is an emergency characterized by 'lead-pipe' rigidity, hyperthermia (high fever), and altered mental status.

115. Treatment for Neuroleptic Malignant Syndrome (NMS) includes immediately stopping the antipsychotic and administering:

- A) Haloperidol
- B) Dantrolene or Bromocriptine**
- C) Lorazepam
- D) Propranolol



Hint: Dantrolene (a muscle relaxant) or Bromocriptine (a dopamine agonist) are the specific antidotes to reverse the fatal rigidity and dopamine blockade.

116. Sedation from antipsychotic drugs is primarily due to the potent blockade of which receptor?

- A) Dopamine D2
- B) Histamine H1**
- C) Serotonin 5-HT2A
- D) Alpha-adrenergic

 **Hint:** Drugs like chlorpromazine, olanzapine, and clozapine strongly block the H1-histamine receptor, which is responsible for their heavy sedative effects.

117. Orthostatic hypotension associated with antipsychotics is caused by the blockade of:

- A) Muscarinic receptors
- B) Alpha-Adrenergic receptors**
- C) Histamine receptors
- D) Dopamine receptors

 **Hint:** Alpha-blockade prevents the blood vessels from constricting when the patient stands up, causing blood to pool in the legs and blood pressure to drop.

118. Significant weight gain and metabolic syndrome are most commonly associated with which class of antipsychotics?

- A) High-potency typicals (Haloperidol)
- B) Atypical neuroleptics (like Olanzapine and Clozapine)**
- C) Barbiturates
- D) All typical neuroleptics

 **Hint:** While Atypicals have a much lower risk of motor side effects (EPS), their major drawback is severe weight gain and metabolic disturbances.

119. Antipsychotics may lower the seizure threshold and should be used with extreme caution or contraindicated in patients with:

A) Hypertension

B) Epilepsy or Seizure disorders

C) Diabetes

D) Asthma

 **Hint:** Drugs like chlorpromazine and clozapine significantly lower the seizure threshold, making them dangerous for patients with pre-existing epilepsy.

120. Levodopa (used for Parkinson's disease) interacts with antipsychotics by:

A) Enhancing the antipsychotic effect

B) Having its effect blocked by the antipsychotic, worsening Parkinson's symptoms

C) Causing Serotonin Syndrome

D) Increasing sedation

 **Hint:** Levodopa increases dopamine, while antipsychotics block dopamine receptors. They directly antagonize each other, rendering the Parkinson's treatment useless.

121. Which of the following is considered a 'First-line' Atypical antipsychotic drug that shows fewer relapses than Haloperidol?

A) Chlorpromazine

B) Risperidone

C) Thioridazine

D) Fluphenazine

 **Hint:** Risperidone is a highly effective first-line atypical agent. It has minimal sedation and fewer relapses, though it can still elevate prolactin levels.

122. Ziprasidone is an atypical neuroleptic that has minimal weight gain but is strictly contraindicated in patients with:

- A) Asthma
- B) A history of cardiac arrhythmias**
- C) Peptic ulcer disease
- D) Renal failure

 **Hint:** Ziprasidone can prolong the QT interval, making it extremely dangerous for anyone with a history of cardiac arrhythmias.

123. A patient taking an antipsychotic presents with dry mouth, blurred vision, urinary retention, and constipation. These are signs of:

- A) Alpha-adrenergic blockade
- B) Antimuscarinic (Anticholinergic) effects**
- C) Histamine H1 blockade
- D) Dopamine supersensitivity

 **Hint:** Drugs like Thioridazine and Olanzapine heavily block muscarinic receptors, causing classic anticholinergic side effects (dry mouth, retention).

124. According to the Doctor's lecture, what is the best initial guide for a physician when choosing a specific antipsychotic drug for a patient?

- A) Always start with the newest atypical drug
- B) The patient's 'past history of response' to a specific drug**
- C) Always start with Haloperidol
- D) The cost of the medication

 **Hint:** The doctor stressed that finding the right antipsychotic is 'trial and error'. If a patient responded well to a specific drug in the past without severe side effects, you **MUST** stick with it.

125. Are neuroleptic drugs curative for schizophrenia?

- A) Yes, they permanently cure the disease
- B) No, they are not curative; they only decrease the intensity of symptoms**
- C) Yes, but only atypical agents cure it
- D) They cure the negative symptoms but not the positive ones

 **Hint:** Neuroleptics do not eliminate the chronic thought disorder. They merely manage the symptoms (hallucinations/delusions) so the patient can function in society.

126. A patient taking Fluphenazine exhibits severe motor restlessness, an inability to sit still, and constant pacing. This extrapyramidal symptom is known as:

- A) Dystonia
- B) Akathisia**
- C) Tardive Dyskinesia
- D) Parkinsonism

 **Hint:** Akathisia is defined as severe motor restlessness and the urge to constantly move, a common EPS seen with high-potency typicals.

127. Promethazine is classified under neuroleptics but is actually a poor antipsychotic. What is its primary clinical use?

- A) Treating bipolar depression
- B) Treating pruritus (itching) due to its strong antihistaminic properties**
- C) Curing schizophrenia
- D) Preventing tardive dyskinesia

 **Hint:** Promethazine is primarily used as a strong antihistamine for allergies, itching, and sometimes nausea, rather than as a primary antipsychotic.

128. Which typical neuroleptics are available as slow-release 'depot' formulations for non-compliant patients?

- A) Clozapine and Risperidone
- B) Fluphenazine and Haloperidol**
- C) Thioridazine and Chlorpromazine
- D) Ziprasidone and Quetiapine

 **Hint:** Haloperidol and Fluphenazine (high potency typicals) can be given as long-acting depot injections (e.g., every 2-4 weeks) for patients who forget to take daily pills.

129. Antipsychotics are NOT effective for treating which type of nausea?

- A) Drug-induced vomiting
- B) Radiation-induced nausea
- C) Motion sickness**
- D) Post-operative nausea

 **Hint:** Motion sickness is mediated by vestibular pathways (histamine/acetylcholine), not dopamine. Scopolamine or antihistamines are used, NOT powerful neuroleptics.

130. Which of the following is NOT an atypical antipsychotic?

- A) Quetiapine
- B) Aripiprazole
- C) Thiothixene**
- D) Paliperidone

 **Hint:** Thiothixene is a high-potency TYPICAL neuroleptic, grouping with Haloperidol and Fluphenazine.

131. Clozapine has a uniquely high affinity for which dopamine receptor subtype, which partially explains its minimal ability to cause EPS?

- A) D1 receptor
- B) D2 receptor
- C) D3 receptor

D) D4 receptor



Hint: Unlike typicals that strongly bind D2, Clozapine binds strongly to D4 and 5-HT2A receptors, providing excellent efficacy with very low EPS risk.

132. Neuroleptics can cause 'Poikilothermia'. What does this mean?

A) The patient constantly feels a burning sensation

B) Body temperature varies directly with the environment because the hypothalamus is depressed

- C) Severe hyperthermia due to muscle breakdown
- D) Irreversible hypothermia



Hint: Neuroleptics depress the temperature-regulating center in the hypothalamus, making the patient's body temperature fluctuate with the room temperature.

133. In schizophrenia, the patient shifting rapidly from one completely unrelated thought to another is a disturbance in the 'form of thoughts' known as:

A) Delusion

B) Loose associations

- C) Hallucination
- D) Blunted affect



Hint: As the doctor explained, 'loose associations' mean the patient's sentences and thoughts are entirely disconnected from one another.

134. If an elderly patient has pre-existing urinary retention or BPH, which typical antipsychotic should be heavily avoided?

- A) Haloperidol
- B) Thioridazine**
- C) Fluphenazine
- D) Thiothixene

 **Hint:** Thioridazine has very strong antimuscarinic (anticholinergic) activity, which will severely worsen urinary retention and constipation.

135. Which atypical antipsychotic is highly notable for treating Bipolar Depression as well as Schizophrenia?

- A) Haloperidol
- B) Quetiapine**
- C) Thioridazine
- D) Chlorpromazine

 **Hint:** Quetiapine (along with Aripiprazole, Olanzapine, and Ziprasidone) is an atypical neuroleptic frequently used to manage the depressive phase of bipolar disorder.

136. Prolonged 'holidays' from neuroleptics (drug holidays) might be attempted specifically to help diminish the symptoms of which adverse effect?

- A) Acute Dystonia
- B) Tardive Dyskinesia**
- C) Akathisia
- D) Neuroleptic Malignant Syndrome

 **Hint:** Tardive Dyskinesia is often irreversible, but stopping the drug for a long period (drug holiday) may allow the supersensitive dopamine receptors to reset and diminish symptoms.

137. The use of Atypical antipsychotics has largely replaced Typical antipsychotics because Atypicals:

- A) Are significantly cheaper
- B) Have a much lower incidence of debilitating extrapyramidal movement disorders**
- C) Do not cause weight gain
- D) Are administered intravenously only

 **Hint:** The primary clinical advantage of 2nd Gen Atypicals is their ability to treat psychosis while heavily minimizing the crippling motor side effects (EPS).

138. Which symptom is a classic presentation of 'Acute Dystonia' resulting from typical antipsychotic use?

- A) Constant pacing back and forth
- B) Sustained, painful contraction of muscles leading to twisting postures (e.g., neck spasms)**
- C) Lip smacking and tongue darting
- D) High fever and rigidity

 **Hint:** Acute dystonia involves sudden, severe muscle spasms (often affecting the face, neck, or back). It is treated quickly with an anticholinergic like Benztropine.

139. Sexual dysfunction, infertility, and amenorrhea in patients taking typical antipsychotics are primarily due to:

- A) Serotonin blockade
- B) Histamine blockade
- C) Elevation of Prolactin due to D2 blockade in the pituitary**
- D) Alpha-adrenergic blockade

 **Hint:** Blocking dopamine in the tuberoinfundibular pathway removes the 'brake' on prolactin, leading to hyperprolactinemia and resulting hormonal/sexual dysfunctions.

140. Which of the following describes 'Anhedonia', a negative symptom of schizophrenia?

- A) Hearing voices
- B) Not getting pleasure from normally pleasurable stimuli**
- C) Believing one is a famous historical figure
- D) Severe motor restlessness

 **Hint:** Anhedonia (inability to feel pleasure), apathy, and blunted affect are negative symptoms that respond very poorly to typical antipsychotics but better to atypicals.

141. Aripiprazole and Thioridazine are exceptions among neuroleptics because they do NOT possess strong:

- A) Antipsychotic effects
- B) Antiemetic effects**
- C) Sedative effects
- D) D2 blocking properties

 **Hint:** Most neuroleptics block the CTZ and act as antiemetics. Aripiprazole and Thioridazine are notable exceptions that do not have this antiemetic property.

142. What happens when an antipsychotic drug blocks the GABA receptors?

- A) It cures schizophrenia
- B) It lowers the seizure threshold**
- C) It causes severe EPS
- D) It induces sleep

 **Hint:** Neuroleptics can block GABA (the main inhibitory neurotransmitter), which lowers the seizure threshold, explaining why they must be used cautiously in epileptic patients.

143. According to the Doctor's lecture, what does 'trial and error' mean in the context of prescribing neuroleptics?

- A) Giving random doses until the patient stops complaining
- B) No one drug is clinically superior; the doctor must try different agents until finding the one that works for that specific patient's profile and side effect tolerance**
- C) Testing typicals only
- D) Using placebo first

 **Hint:** Because patients have highly variable responses and sensitivities to side effects, doctors must 'switch' drugs until finding the optimal match for the individual.

144. Which of the following Atypical antipsychotics is frequently used in the treatment of Autism?

- A) Clozapine
- B) Ziprasidone
- C) Risperidone**
- D) Fluphenazine

 **Hint:** Risperidone is widely used to manage irritability, aggression, and mood swings in patients with Autism, alongside its use in schizophrenia and bipolar depression.

145. The primary advantage of Atypical neuroleptics over Typical ones is their ability to:

- A) Be administered once a year
- B) Treat negative symptoms and lower EPS risk**
- C) Avoid weight gain completely
- D) Cure the disease permanently

 **Hint:** Atypicals are defined by their low EPS profile and their superior efficacy in treating the socially debilitating 'negative' symptoms of schizophrenia.

146. If a patient on an antipsychotic develops 'Poikilothermia', the nurse should advise them to:

A) Stop the medication immediately

B) Avoid extreme heat or cold environments because their body cannot regulate temperature properly

C) Take an antipyretic

D) Drink more water



Hint: Because the hypothalamus is depressed, the patient loses the ability to sweat or shiver effectively, making them vulnerable to environmental temperature extremes.

147. Which of the following side effects is MOST likely to cause a patient to be non-compliant with Atypical antipsychotic therapy (like Olanzapine)?

A) Severe weight gain and metabolic syndrome

B) Parkinson-like tremors

C) Ringing in the ears

D) Dental caries



Hint: While Atypicals spare the patient from motor tremors, they cause massive weight gain, high cholesterol, and diabetes risk, which often leads patients to stop taking them.

148. Which of the following typical neuroleptics is considered 'Low Potency'?

A) Haloperidol

B) Fluphenazine

C) Chlorpromazine

D) Pimozide



Hint: Chlorpromazine and Thioridazine are low-potency typicals. They require higher doses and have higher sedative/anticholinergic effects but lower EPS compared to high-potency ones like Haloperidol.

149. A patient arrives at the ER with a fixed, distorted posture, twisting of the neck, and sustained muscle contractions after starting a neuroleptic. The immediate treatment should be:

A) Levodopa

B) Benztropine (Anticholinergic)

C) Naloxone

D) Dantrolene



Hint: This is Acute Dystonia (an EPS). It is caused by an excess of acetylcholine relative to dopamine, so it is rapidly reversed by an anticholinergic drug.

150. Which of the following statements best describes the mechanism of action of typical neuroleptics?

A) They increase dopamine release in the cerebral cortex

B) They are competitive inhibitors at dopamine D2 receptors

C) They block serotonin reuptake

D) They stimulate GABA receptors



Hint: The fundamental mechanism of the first-generation (typical) antipsychotic drugs is the competitive blockade of D2 dopamine receptors.